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Partition of a set

In mathematics, a **partition of a set** is a grouping of the set's elements into non-empty subsets, in such a way that every element is included in one and only one of the subsets.



A set of stamps partitioned into bundles: No stamp is in two bundles, no bundle is empty, and every stamp is in a bundle.

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Definition

A partition of a set X is a set of nonempty subsets of X such that every element x in X is in exactly one of these subsets[2] (i.e., X is a disjoint union of the subsets).

Equivalently, a family of sets P is a partition of X if and only if all of the following conditions hold:[3]

- The family P does not contain the empty set (that is $\emptyset \notin P$).
- The union of the sets in P is equal to X (that is $\bigcup_{A \in P} A = X$). The sets in P are said to **cover** X.
- The intersection of any two distinct sets in P is empty (that is $(\forall A, B \in P)\ A \neq B \implies A \cap B = \emptyset$). The elements of P are said to be pairwise disjoint.

The sets in P are called the *blocks*, *parts* or *cells* of the partition.[4]

The **rank** of P is $|X| - |P|$, if X is finite.

Examples

- Every singleton set $\{x\}$ has exactly one partition, namely $\{\{x\}\}$.
- The empty set $\emptyset$ has exactly one partition, namely $\emptyset$.
- For any nonempty set X, $P = \{X\}$ is a partition of X, called the **trivial partition**.
- For any non-empty proper subset A of a set U, the set A together with its complement form a partition of U, namely, $\{A, U \setminus A\}$.
- The set { 1, 2, 3 } has these five partitions (one partition per item):
  - { {1}, {2}, {3} }, sometimes written 1|2|3.
  - { {1, 2}, {3} }, or 12|3.
  - { {1, 3}, {2} }, or 13|2.
  - { {1}, {2, 3} }, or 1|23.

- { {1, 2, 3} }, or 123 (in contexts where there will be no confusion with the number).

- The following are not partitions of { 1, 2, 3 }:
  - { {}, {1, 3}, {2} } is not a partition (of any set) because one of its elements is the empty set.
  - { {1, 2}, {2, 3} } is not a partition (of any set) because the element 2 is contained in more than one block.
  - { {1}, {2} } is not a partition of {1, 2, 3} because none of its blocks contains 3; however, it is a partition of {1, 2}.

Partitions and equivalence relations

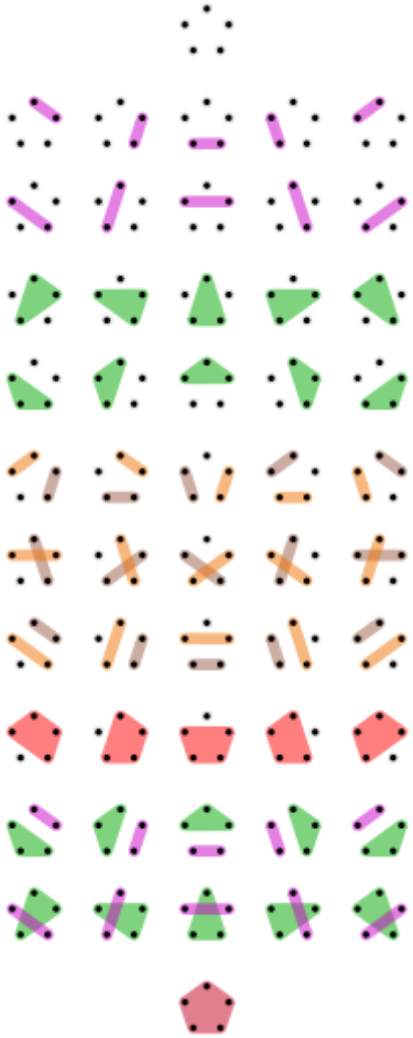
For any equivalence relation on a set X, the set of its equivalence classes is a partition of X. Conversely, from any partition P of X, we can define an equivalence relation on X by setting $x \sim y$ precisely when x and y are in the same part in P. Thus the notions of equivalence relation and partition are essentially equivalent.[5]

The axiom of choice guarantees for any partition of a set X the existence of a subset of X containing exactly one element from each part of the partition. This implies that given an equivalence relation on a set one can select a canonical representative element from every equivalence class.

Refinement of partitions

A partition α of a set X is a **refinement** of a partition ρ of X—and we say that α is *finer* than ρ and that ρ is *coarser* than α—if every element of α is a subset of some element of ρ. Informally, this means that α is a further fragmentation of ρ. In that case, it is written that $\alpha \leq \rho$.

This *finer-than* relation on the set of partitions of X is a partial order (so the notation "≤" is appropriate). Each set of elements has a least upper bound and a greatest lower bound, so that it forms a lattice, and more specifically (for partitions of a finite set) it is a geometric lattice.[6] The *partition lattice* of a 4-element set has 15 elements and is depicted in the Hasse diagram on the left.

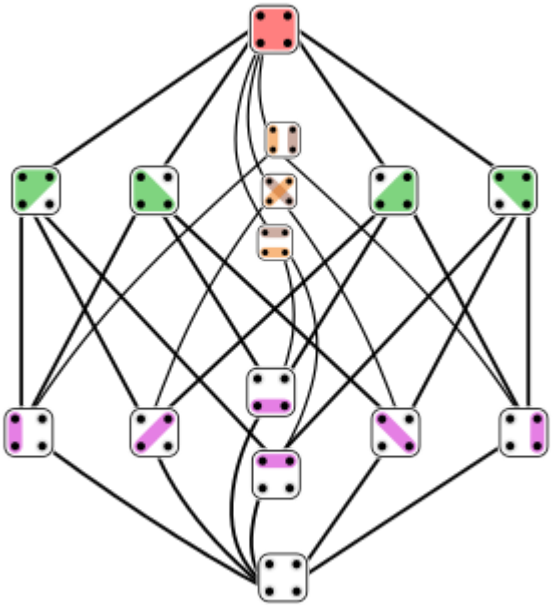


The 52 partitions of a set with 5 elements. A colored region indicates a subset of X, forming a member of the enclosing partition. Uncolored dots indicate single-element subsets. The first shown partition contains five single-element subsets; the last partition contains one subset having five elements.

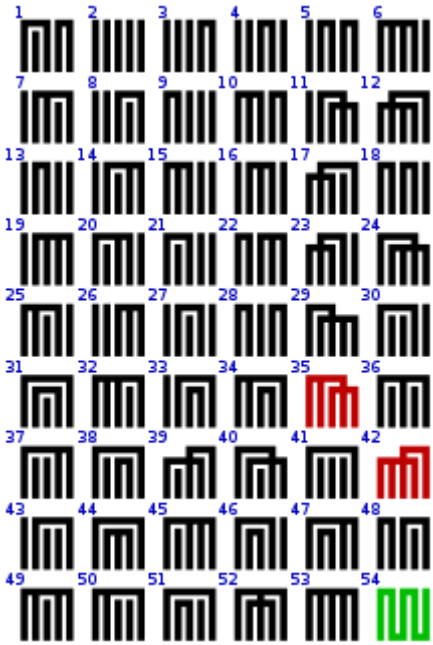
Based on the cryptomorphism between geometric lattices and matroids, this lattice of partitions of a finite set corresponds to a matroid in which the base set of the matroid consists of the atoms of the lattice, namely, the partitions with $n - 2$ singleton sets and one two-element set. These atomic partitions correspond one-for-one with the edges of a complete graph. The matroid closure of a set of atomic partitions is the finest common coarsening of them all; in graph-theoretic terms, it is the partition of the vertices of the complete graph into the connected components of the subgraph formed by the given set of edges. In this way, the lattice of partitions corresponds to the lattice of flats of the graphic matroid of the complete graph.

Another example illustrates the refining of partitions from the perspective of equivalence relations. If D is the set of cards in a standard 52-card deck, the *same-color-as* relation on D – which can be denoted $\sim_C$ – has two equivalence classes: the sets {red cards} and {black cards}. The 2-part partition corresponding to $\sim_C$ has a refinement that yields the *same-suit-as* relation $\sim_S$, which has the four equivalence classes {spades}, {diamonds}, {hearts}, and {clubs}.

Noncross partition



Partitions of a 4-set ordered by refinement



The traditional Japanese symbols for the chapters of the *Tale of Genji* are based on the 52 ways of partitioning five elements.[1]

A partition of the set $N = \{1, 2, ..., n\}$ with corresponding equivalence relation ~ is **noncrossing** if it has the following property: If four elements a, b, c and d of N having $a < b < c < d$ satisfy $a \sim c$ and $b \sim d$, then $a \sim b \sim c \sim d$. The name comes from the following equivalent definition: Imagine the elements 1, 2, ..., n of N drawn as the n vertices of a regular n-gon (in counterclockwise order). A partition can then be visualized by drawing each block as a polygon (whose vertices are the elements of the block). The partition is then noncrossing if and only if these polygons do not intersect.

The lattice of noncrossing partitions of a finite set has recently taken on importance because of its role in free probability theory. These form a subset of the lattice of all partitions, but not a sublattice, since the join operations of the two lattices do not agree.

Counting partitions

The total number of partitions of an n-element set is the Bell number B_n. The first several Bell numbers are $B_0 = 1$, $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$, and $B_6 = 203$ (sequence A000110 in the OEIS). Bell numbers satisfy the recursion

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$$

and have the exponential generating function

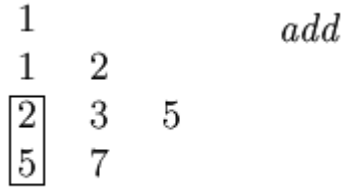
$$\sum_{n=0}^{\infty} \frac{B_n}{n!} z^n = e^{e^z - 1}.$$

The Bell numbers may also be computed using the Bell triangle in which the first value in each row is copied from the end of the previous row, and subsequent values are computed by adding two numbers, the number to the left and the number to the above left of the position. The Bell numbers are repeated along both sides of this triangle. The numbers within the triangle count partitions in which a given element is the largest singleton.

The number of partitions of an n-element set into exactly k nonempty parts is the Stirling number of the second kind $S(n, k)$.

The number of noncrossing partitions of an n-element set is the Catalan number C_n, given by

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$



Construction of the Bell triangle

See also

- Exact cover
- Cluster analysis
- Weak ordering (ordered set partition)
- Equivalence relation
- Partial equivalence relation
- Partition refinement
- List of partition topics
- Lamination (topology)
- Rhyme schemes by set partition

Notes

1. Knuth, Donald E. (2013), "Two thousand years of combinatorics", in Wilson, Robin; Watkins, John J., *Combinatorics: Ancient and Modern*, Oxford University Press, pp. 7–37
2. Naive Set Theory (1960). *Halmos, Paul R.* (https://books.google.com/books?id=x6cZBQ9qtgoC&pg=PA28) Springer. p. 28. ISBN 9780387900926.
3. Lucas, John F. (1990). *Introduction to Abstract Mathematics* (https://books.google.com/books?id=jklsb5JUgoQC&pg=PA187). Rowman & Littlefield. p. 187. ISBN 9780912675732.
4. Brualdi, *pp*. 44–45
5. Schechter, *p*. 54
6. Birkhoff, Garrett (1995), *Lattice Theory* (https://books.google.com/books?id=0Y8d-MdtVwkC&pg=PA95), Colloquium Publications, **25** (3rd ed.), American Mathematical Society, p. 95, ISBN 9780821810255.

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- Brualdi, Richard A. (2004). *Introductory Combinatorics* (4th ed.). Pearson Prentice Hall. ISBN 0-13-100119-1.
- Schechter, Eric (1997). *Handbook of Analysis and Its Foundations*. Academic Press. ISBN 0-12-622760-8.

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